A. SETUP ENVIRONMENT

Create a Table with sample records

```
USE db_query_builder;


CREATE TABLE employees (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100) NOT NULL,
    email VARCHAR(150) UNIQUE NOT NULL,
    phone VARCHAR(50) NULL,
    position VARCHAR(100) NULL,
    department VARCHAR(100) NULL,
    salary DECIMAL(10,2) DEFAULT 0.00,
    hire_date DATE NOT NULL,
    status ENUM('active', 'inactive', 'terminated') DEFAULT 'active',
    created_at TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP
);

INSERT INTO employees (first_name, last_name, email, phone, position,
department, salary, hire_date, status) VALUES
('John', 'Smith', 'john.smith@example.com', '09171234567', 'Software
Engineer', 'IT', 60000.00, '2020-03-15', 'active'),
('Maria', 'Garcia', 'maria.garcia@example.com', '09281234568', 'HR Officer',
'Human Resources', 45000.00, '2019-07-01', 'active'),
('David', 'Lopez', 'david.lopez@example.com', '09181234569', 'Sales
Executive', 'Sales', 52000.00, '2021-04-23', 'active'),
('Jessica', 'Cruz', 'jessica.cruz@example.com', '09171234570', 'Accountant',
'Finance', 48000.00, '2020-09-18', 'active'),
('Robert', 'Tan', 'robert.tan@example.com', '09351234571', 'IT Support',
'IT', 38000.00, '2022-01-10', 'active'),
('Angela', 'Reyes', 'angela.reyes@example.com', '09451234572', 'Marketing
Specialist', 'Marketing', 50000.00, '2020-06-21', 'active'),
('Michael', 'Chua', 'michael.chua@example.com', '09191234573', 'Systems
Analyst', 'IT', 65000.00, '2018-05-03', 'active'),
('Patricia', 'Lim', 'patricia.lim@example.com', '09221234574', 'Recruitment
Officer', 'Human Resources', 42000.00, '2021-11-08', 'active'),
('Kevin', 'Dela Cruz', 'kevin.delacruz@example.com', '09181234575', 'Finance
Analyst', 'Finance', 55000.00, '2022-02-15', 'active'),
('Catherine', 'Santos', 'catherine.santos@example.com', '09171234576', 'Sales
Assistant', 'Sales', 35000.00, '2023-03-01', 'active'),
('Paul', 'Mendoza', 'paul.mendoza@example.com', '09381234577', 'Software
Developer', 'IT', 62000.00, '2020-08-25', 'active'),
('Grace', 'Ocampo', 'grace.ocampo@example.com', '09291234578', 'HR
Assistant', 'Human Resources', 37000.00, '2021-09-09', 'active'),
('Jason', 'Torres', 'jason.torres@example.com', '09161234579', 'Marketing
Officer', 'Marketing', 48000.00, '2019-05-14', 'active'),
('Emily', 'Castro', 'emily.castro@example.com', '09201234580', 'Sales
Representative', 'Sales', 40000.00, '2021-01-19', 'active'),
('Steven', 'Uy', 'steven.uy@example.com', '09391234581', 'Network
Administrator', 'IT', 58000.00, '2020-02-10', 'active'),
```

```
('Monica', 'Villanueva', 'monica.villanueva@example.com', '09181234582',
'Finance Officer', 'Finance', 49000.00, '2022-03-12', 'active'),
('Nathan', 'Bautista', 'nathan.bautista@example.com', '09151234583', 'IT
Technician', 'IT', 40000.00, '2023-06-01', 'active'),
('Andrea', 'Go', 'andrea.go@example.com', '09481234584', 'HR Specialist',
'Human Resources', 47000.00, '2020-07-30', 'active'),
('Mark', 'Villamor', 'mark.villamor@example.com', '09191234585', 'Sales
Manager', 'Sales', 72000.00, '2018-09-22', 'active'),
('Diana', 'Ramos', 'diana.ramos@example.com', '09281234586', 'Marketing
Manager', 'Marketing', 70000.00, '2019-10-10', 'active'),
('Patrick', 'Santiago', 'patrick.santiago@example.com', '09181234587',
'Senior Developer', 'IT', 78000.00, '2017-11-05', 'active'),
('Christine', 'David', 'christine.david@example.com', '09291234588', 'HR
Supervisor', 'Human Resources', 55000.00, '2018-12-12', 'active'),
('Raymond', 'Flores', 'raymond.flores@example.com', '09161234589', 'Finance
Manager', 'Finance', 75000.00, '2016-08-08', 'active'),
('Elaine', 'Tan', 'elaine.tan@example.com', '09451234590', 'Sales
Coordinator', 'Sales', 38000.00, '2021-10-01', 'active'),
('Leo', 'Castillo', 'leo.castillo@example.com', '09171234591', 'Graphic
Designer', 'Marketing', 46000.00, '2020-05-06', 'active'),
('Theresa', 'Lim', 'theresa.lim@example.com', '09281234592', 'Software
Tester', 'IT', 53000.00, '2021-12-20', 'active'),
('Brian', 'Reyes', 'brian.reyes@example.com', '09181234593', 'Account
Executive', 'Sales', 49000.00, '2022-04-02', 'active'),
('Isabelle', 'Morales', 'isabelle.morales@example.com', '09191234594',
'Recruiter', 'Human Resources', 43000.00, '2020-11-09', 'active'),
('Felix', 'Gonzales', 'felix.gonzales@example.com', '09381234595', 'IT
Administrator', 'IT', 60000.00, '2019-02-16', 'active'),
('Nina', 'Fernandez', 'nina.fernandez@example.com', '09151234596', 'Payroll
Officer', 'Finance', 48000.00, '2020-04-05', 'active'),
('Albert', 'Diaz', 'albert.diaz@example.com', '09451234597', 'Marketing
Coordinator', 'Marketing', 45000.00, '2023-01-10', 'active'),
('Veronica', 'Santos', 'veronica.santos@example.com', '09171234598', 'Sales
Support', 'Sales', 36000.00, '2021-06-21', 'active'),
('Jerome', 'Delos Reyes', 'jerome.delosreyes@example.com', '09291234599',
'Backend Developer', 'IT', 64000.00, '2022-09-18', 'active'),
('Samantha', 'Lee', 'samantha.lee@example.com', '09381234600', 'HR Analyst',
'Human Resources', 44000.00, '2020-08-08', 'active'),
('Arthur', 'Cruz', 'arthur.cruz@example.com', '09161234601', 'Finance Clerk',
'Finance', 37000.00, '2023-05-10', 'active'),
('Melissa', 'Torres', 'melissa.torres@example.com', '09181234602', 'Sales
Officer', 'Sales', 42000.00, '2019-09-19', 'active'),
('George', 'Lim', 'george.lim@example.com', '09191234603', 'IT Support
Specialist', 'IT', 46000.00, '2021-03-22', 'active'),
('Hannah', 'Uy', 'hannah.uy@example.com', '09481234604', 'Marketing
Assistant', 'Marketing', 40000.00, '2020-02-14', 'active'),
('Christian', 'Reyes', 'christian.reyes@example.com', '09281234605', 'Senior
Analyst', 'Finance', 72000.00, '2018-06-07', 'active'),
('Jasmine', 'Ramos', 'jasmine.ramos@example.com', '09171234606', 'Sales
Associate', 'Sales', 34000.00, '2023-02-02', 'active'),
('Erik', 'Chavez', 'erik.chavez@example.com', '09191234607', 'Web Developer',
'IT', 61000.00, '2020-11-11', 'active'),
('Rica', 'Villanueva', 'rica.villanueva@example.com', '09201234608', 'Admin
Assistant', 'HR', 32000.00, '2023-08-15', 'active'),
('Leo', 'Aquino', 'leo.aquino@example.com', '09451234609', 'System Engineer',
'IT', 67000.00, '2019-05-25', 'active'),
```

```
('Ivy', 'Serrano', 'ivy.serrano@example.com', '09181234610', 'HR Clerk',
'Human Resources', 33000.00, '2021-04-28', 'active'),
('Randy', 'Domingo', 'randy.domingo@example.com', '09171234611', 'Finance
Specialist', 'Finance', 52000.00, '2022-06-17', 'active'),
('Paula', 'Lopez', 'paula.lopez@example.com', '09181234612', 'Marketing
Officer', 'Marketing', 47000.00, '2020-09-01', 'active'),
('Victor', 'Manalo', 'victor.manalo@example.com', '09291234613', 'Sales
Executive', 'Sales', 54000.00, '2018-07-29', 'active'),
('Olivia', 'Gutierrez', 'olivia.gutierrez@example.com', '09381234614',
'Software Engineer', 'IT', 68000.00, '2022-10-18', 'active'),
('Henry', 'Ramos', 'henry.ramos@example.com', '09161234615', 'Finance
Analyst', 'Finance', 51000.00, '2021-12-30', 'active'),
('Sofia', 'Cruz', 'sofia.cruz@example.com', '09201234616', 'Marketing
Specialist', 'Marketing', 48000.00, '2019-11-21', 'active');
```

B. SETUP INTERFACE

- Setup new laravel project: laravel new db_query_builder
- Create controller name: php artisan make:controller EmployeesController
- Update your EmployeesController create a function index

EmployeesController.php

app > Http > Controllers > EmployeesController.php > ...

```php
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;

class EmployeesController extends Controller
{
    public function index()
    {

        return '<h1>Hello</h1>';
    }
}
```

- Update web.php create route for EmployessController

web.php

routes > web.php > ...

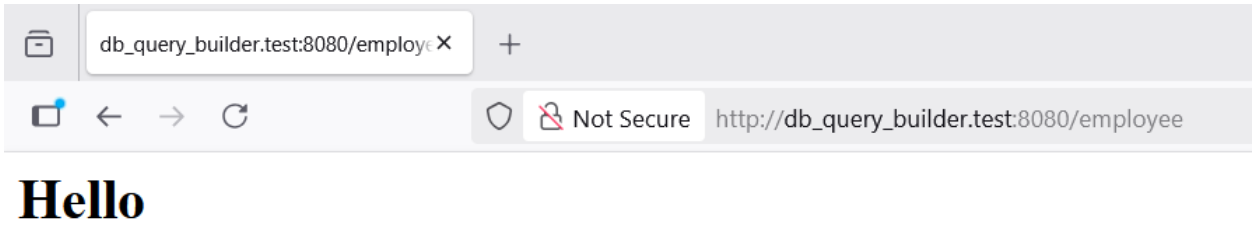
```php
<?php

use App\Http\Controllers\EmployeesController;
use Illuminate\Support\Facades\Route;

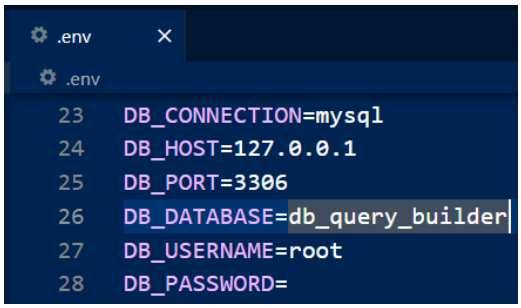
Route::get('/', function () {
    return view('welcome');
});


Route::get('/employee',[EmployeesController::class,'index']);
```

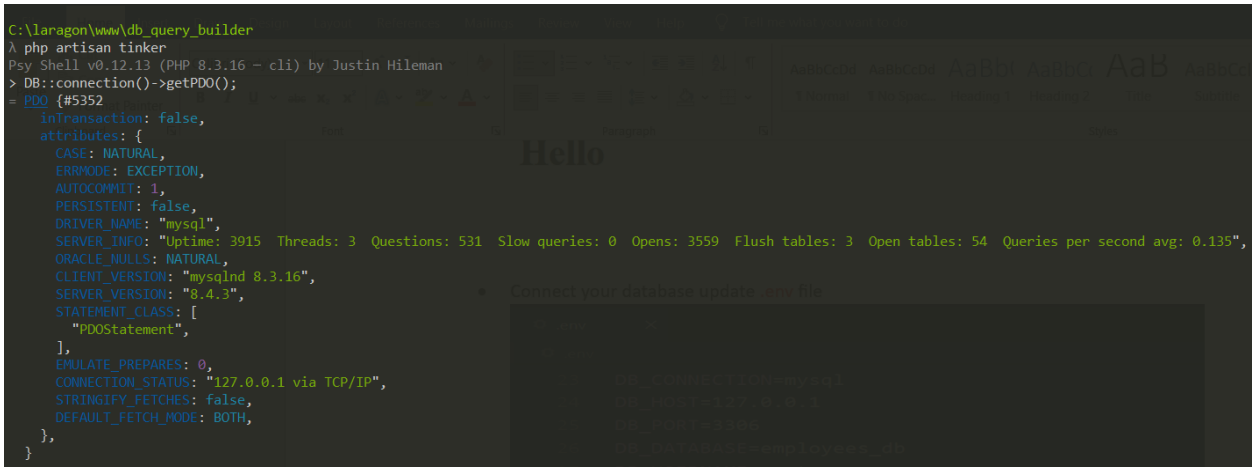
- Test your route



- Connect your database update .env file



- Test your connection in Terminal



C. RETRIEVING DATA FROM TABLE

- Update Employees Controller, to return all records as array

EmployeesController.php

app > Http > Controllers > EmployeesController.php > ...

```php
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;
use Illuminate\Support\facades\DB; //use this to activate DB class

use function Laravel\Prompts\table;

class EmployeesController extends Controller
{
    public function index()
    {
        return DB::table('employees')->get(); //returns all records as array

    }
}
```

Result:

db_query_builder.test:8080/employe

http://db_query_builder.test:8080/employee

JSON Raw Data Headers

Save Copy Collapse All Expand All Filter JSON


```
0:
    id:           1
    first_name:   "John"
    last_name:    "Smith"
    email:        "john.smith@example.com"
    phone:        "09171234567"
    position:     "Software Engineer"
    department:   "IT"
    salary:       "60000.00"
    hire_date:    "2020-03-15"
    status:       "active"
    created_at:   "2025-10-26 18:30:30"
    updated_at:   "2025-10-26 18:30:30"
1:
    id:           2
    first_name:   "Maria"
    last_name:    "Garcia"
    email:        "maria.garcia@example.com"
    phone:        "09281234568"
    position:     "HR Officer"
    department:   "Human Resources"
    salary:       "45000.00"
```

- Update Employees Controller, to return single record base on id as primary key

EmployeesController.php

app > Http > Controllers > EmployeesController.php > ...

```php
<?php

namespace App\Http\Controllers;

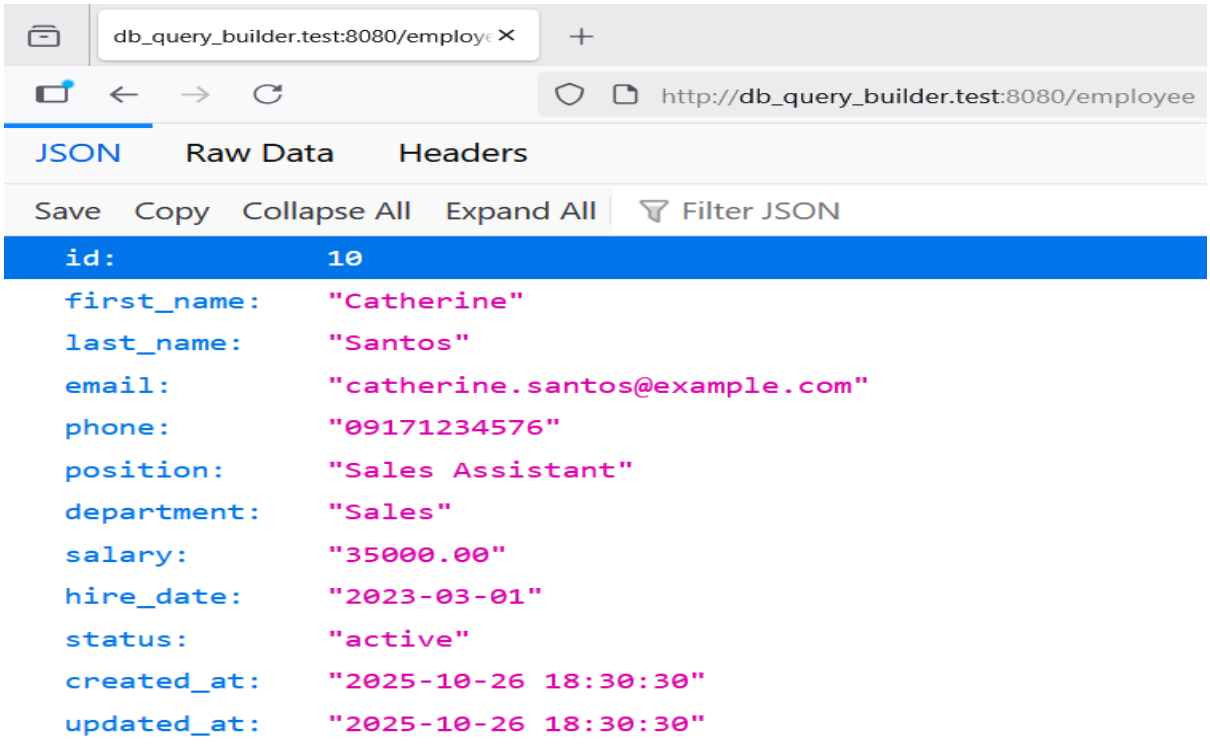
use Illuminate\Http\Request;
use Illuminate\Support\facades\DB; //use this to activate DB class

use function Laravel\Prompts\table;

class EmployeesController extends Controller
{
    public function index()
    {
        // return DB::table('employees')->get(); //returns all records as array
         return DB::table('employees')->find(10);// return a single record based on id as primary key

    }
```

Result:



- Update Employees Controller, to return the first record

EmployeesController.php

app > Http > Controllers > EmployeesController.php > ...

```php
<?php

namespace App\Http\Controllers;

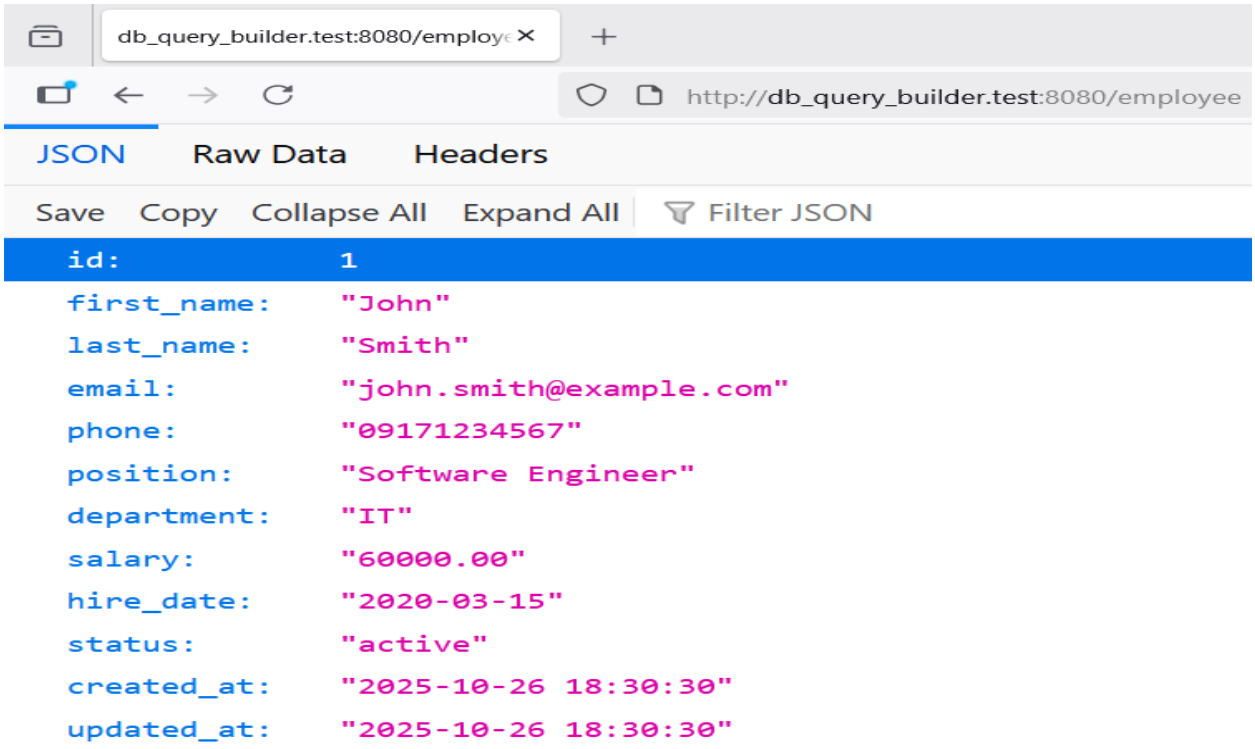
use Illuminate\Http\Request;
use Illuminate\Support\facades\DB; //use this to activate DB class

use function Laravel\Prompts\table;

class EmployeesController extends Controller
{
    public function index()
    {
        // return DB::table('employees')->get(); //returns all records as array
        // return DB::table('employees')->find(10);// return a single record based on id as primary key
            return DB::table('employees')->first(); //return the first record

    }
}
```

Result:



D. RETRIEVING LIST COLUMN VALUES using pluck function(in Laravel, the pluck() method is primarily designed to retrieve a single column's value)

EmployeesController.php

app > Http > Controllers > EmployeesController.php > ...

```php
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;
use Illuminate\Support\facades\DB; //use this to activate DB class

use function Laravel\Prompts\table;

class EmployeesController extends Controller
{
    public function index()
    {
    //  return DB::table('employees')->pluck('first_name'); //return columns with arrary keys index
        return DB::table('employees')->pluck('first_name','id'); //return columns with id as primary key



    }
}
```

Result:

db_query_builder.test:8080/employ

http://db_query_builder.test:8080/employee

JSON Raw Data Headers

Save Copy Collapse All Expand All Filter JSON

```
0:  "John"
1:  "Maria"
2:  "David"
3:  "Jessica"
4:  "Robert"
5:  "Angela"
6:  "Michael"
7:  "Patricia"
8:  "Kevin"
9:  "Catherine"
10: "Paul"
```

db_query_builder.test:8080/employ

http://db_query_builder.test:8080/employee

JSON Raw Data Headers

Save Copy Collapse All Expand All Filter JSON

```
1:  "John"
2:  "Maria"
3:  "David"
4:  "Jessica"
5:  "Robert"
6:  "Angela"
7:  "Michael"
8:  "Patricia"
9:  "Kevin"
10: "Catherine"
```

- If you want to display or retrieve multiple columns instead of pluck use select

EmployeesController.php

app > Http > Controllers > EmployeesController.php > ...

```php
<?php

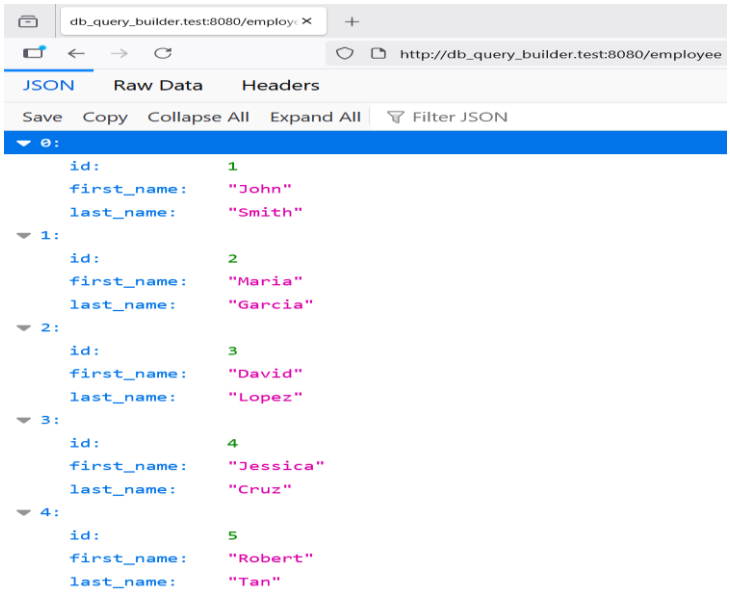
namespace App\Http\Controllers;

use Illuminate\Http\Request;
use Illuminate\Support\facades\DB; //use this to activate DB class

use function Laravel\Prompts\table;

class EmployeesController extends Controller
{
    public function index()
    {
     $employees = DB::table('employees')
                    ->select('id', 'first_name', 'last_name')->get();
     return $employees;
    }
}
```

Result:



E. QUERY BUILDER THE WHERE CONDITION TO RETRIEVE RECORDS

-

EmployeesController.php

app > Http > Controllers > EmployeesController.php > ...

```php
<?php
namespace App\Http\Controllers;
use Illuminate\Http\Request;
use Illuminate\Support\facades\DB; //use this to activate DB class

use function Laravel\Prompts\table;

class EmployeesController extends Controller
{
    public function index()
    {

     //Filter Records using  WHERE

     //$employee =  DB::table('employees')->where('id','=','15')->get();
     //$employee =  DB::table('employees')->where('id','15')->get();
     //$employee =  DB::table('employees')->where('id','>=','15')->get();
     //$employee =  DB::table('employees')->where('id','>=','15')->where('id','<=','30')->get();

     //All records

     //$employee = DB::table('employees')->get();
     //$employee = DB::table('employees')->orderBy('last_name')->get();
       $employee = DB::table('employees')->orderBy('last_name')->orderBy('first_name')->get();

     return $employee;
    }
}
```

EmployeesController.php

app > Http > Controllers > EmployeesController.php > ...

```php
<?php
namespace App\Http\Controllers;
use Illuminate\Http\Request;
use Illuminate\Support\facades\DB; //use this to activate DB class

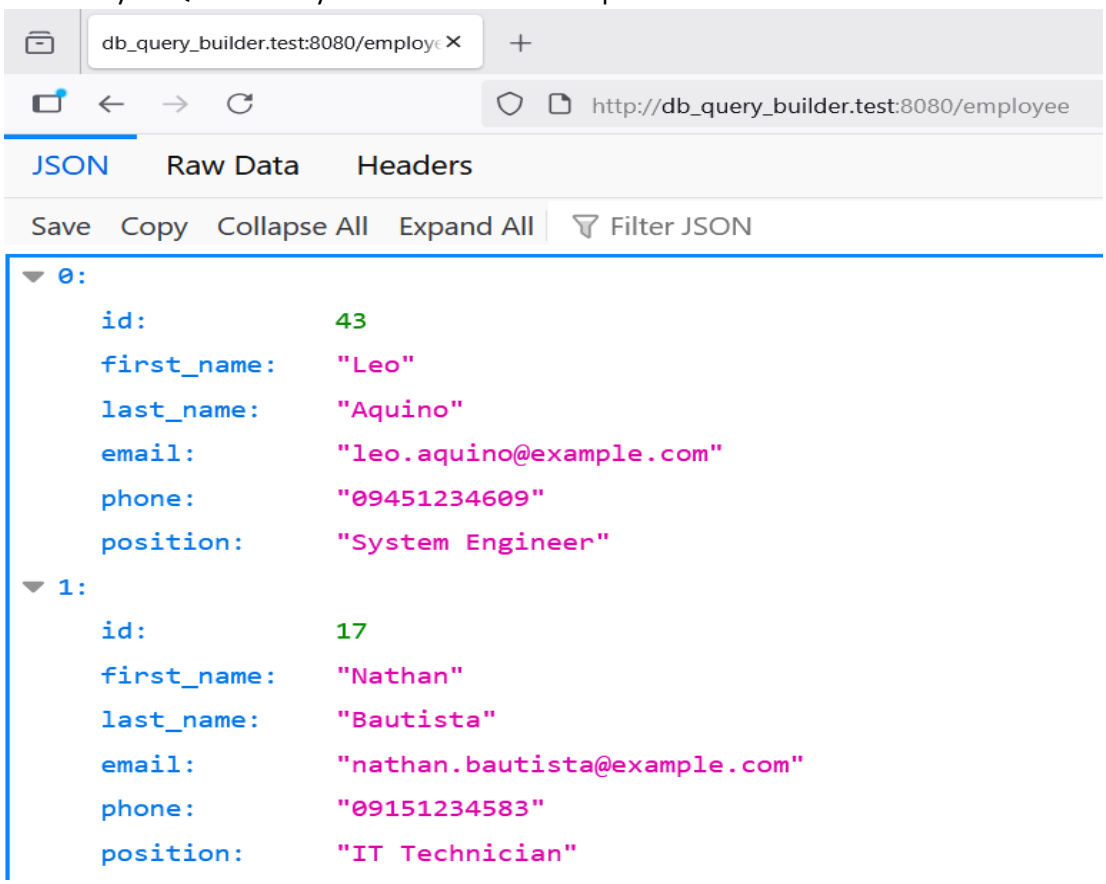
use function Laravel\Prompts\table;

class EmployeesController extends Controller
{
    public function index()
    {

      //All Records with field Selection
       $employee = DB::table('employees')
        ->select('id', 'first_name', 'last_name', 'email', 'phone', 'position')
        ->orderBy('last_name', 'asc')
        ->orderBy('first_name', 'asc')
        ->get();

     return $employee;
    }
}
```

Result: Try all Queries call your route on URL Example



F. QUERY BUILDER INSERT

- **Update controller EmployeesController**

EmployeesController.php

app > Http > Controllers > EmployeesController.php > EmployeesController > index

```php
<?php
namespace App\Http\Controllers;
use Illuminate\Http\Request;
use Illuminate\Support\Facades\DB; //use this to activate DB class

use function Laravel\Prompts\table;

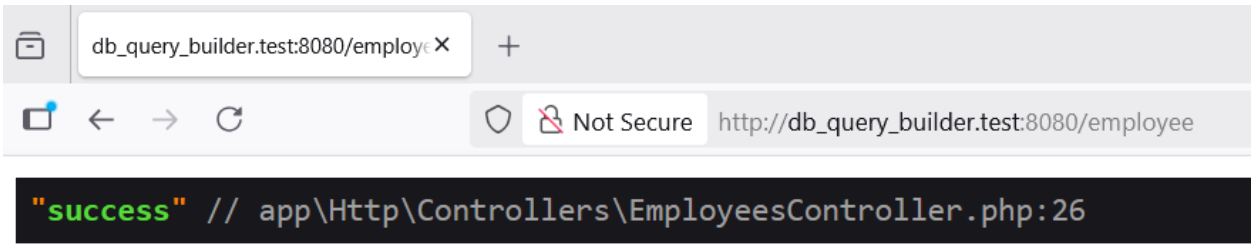
class EmployeesController extends Controller
{
    public function index()
    {

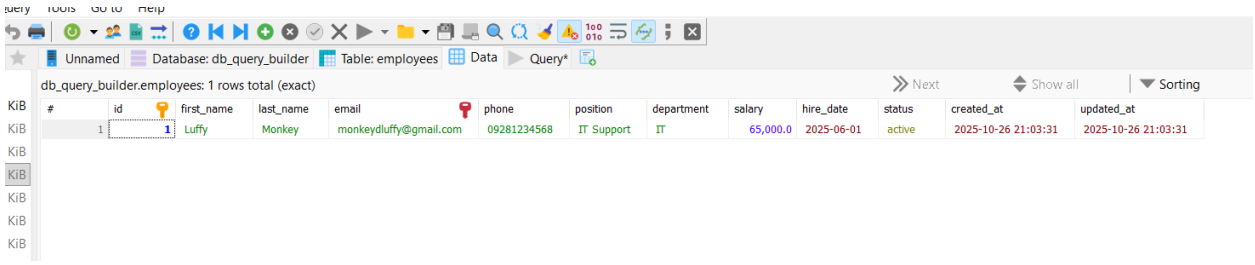
      //INSERTING RECORDS
      DB::table('employees')->insert([
          'first_name' => 'Luffy',
          'last_name' => 'Monkey',
          'email' => 'monkeydluffy@gmail.com',
          'phone' => '09281234568',
          'position' => 'IT Support',
          'department' => 'IT',
          'salary' => '65000.00',
          'hire_date' => '2025-06-01',
          'status' => 'active',
      ]);

         dd('success');

    }
}
```

Result:



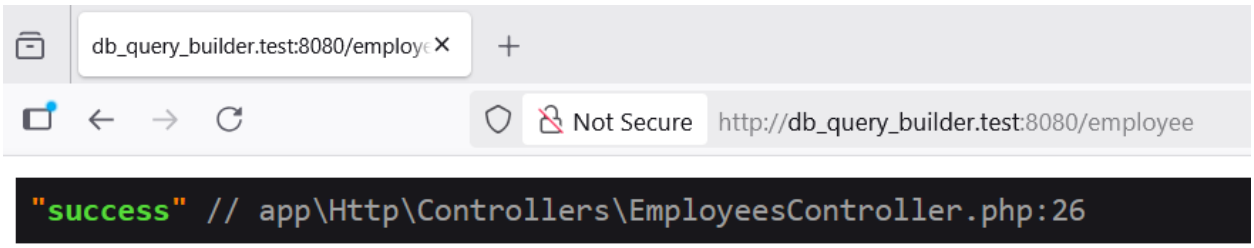


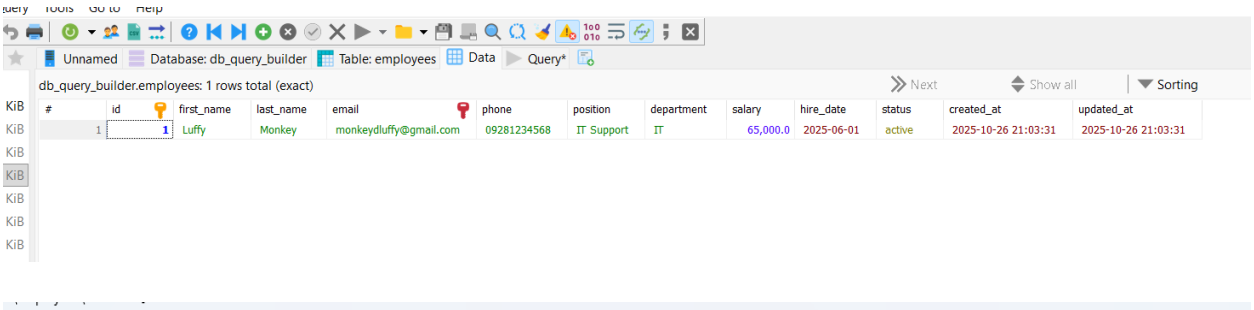
G. QUERY BUILDER UPDATE

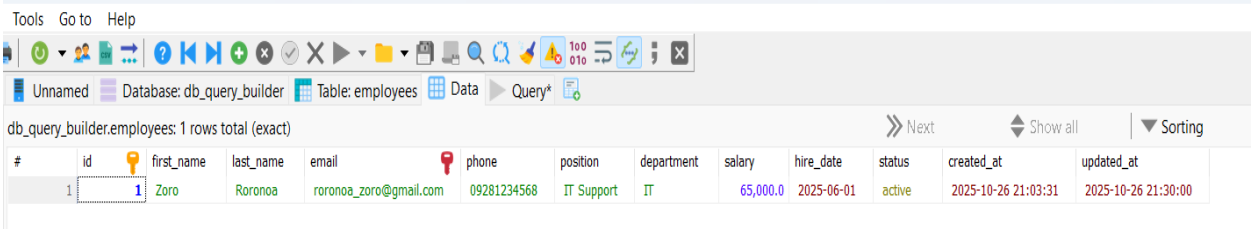
- Update controller EmployeesController.php



Result:

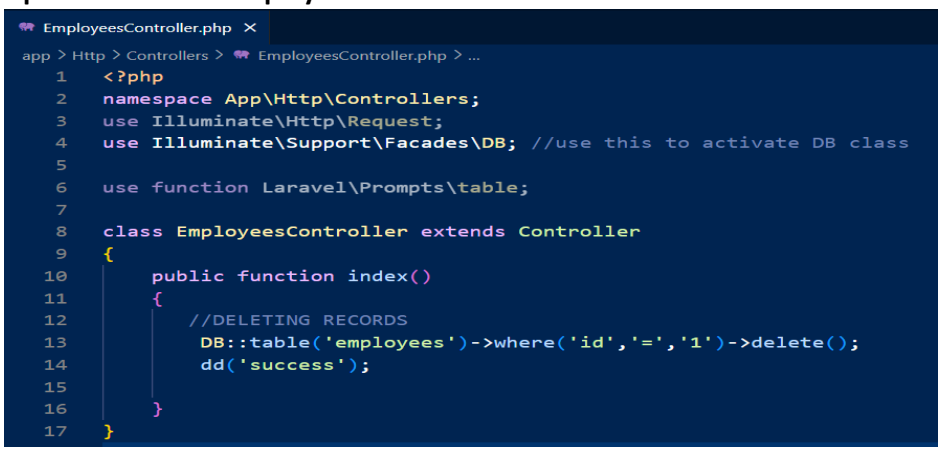






H. QUERY BUILDER DELETE

- **Update controller EmployeesController**



Result: Record Deleted on table where id is equal to 1 as condition

I. QUERY BUILDER JOIN TABLE

- Create new table departments

```
CREATE TABLE departments (
    id INT AUTO_INCREMENT PRIMARY KEY,
    department VARCHAR(100) NOT NULL
);
```

- Insert records

```
INSERT INTO departments (department) VALUES
('Human Resources'),
('Finance'),
('IT'),
('Marketing'),
('Sales'),
('Operations');
```

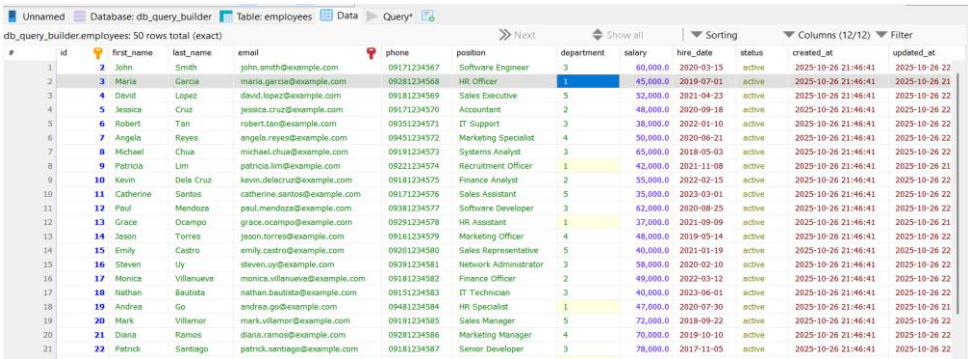
- Update your table employees for department column base on departments id

db_query_builder.departments: 6 rows total (exact)

#	id	department
1	1	Human Resources
2	2	Finance
3	3	IT
4	4	Marketing
5	5	Sales
6	6	Operations

```
UPDATE employees SET department = 1 WHERE department='Human Resources';
UPDATE employees SET department = 2 WHERE department='Finance';
UPDATE employees SET department = 3 WHERE department='IT';
UPDATE employees SET department = 4 WHERE department='Marketing';
UPDATE employees SET department = 5 WHERE department='Sales';
UPDATE employees SET department = 6 WHERE department='Operations';
```

Results on database table



- Update controller EmployeesController.php

EmployeesController.php

app > Http > Controllers > EmployeesController.php > ...

```php
<?php
namespace App\Http\Controllers;
use Illuminate\Http\Request;
use Illuminate\Support\Facades\DB; //use this to activate DB class

use function Laravel\Prompts\table;

class EmployeesController extends Controller
{
    public function index()
    {

      //Show normal table without join
      // $employee = DB::table('employees')->get();

      //Join table
      //$employee =  DB::table('employees')->join('departments','employees.department','=','departments.id')->get();

        //Join table with selected columns
        $employee =  DB::table('employees')->join('departments','employees.department','=','departments.id')
        ->select('employees.last_name','employees.first_name','departments.department')
        ->get();
        return $employee;

    }
}
```

Result:

db_query_builder.test:8080/employe

http://db_query_builder.test:8080/employee

JSON Raw Data Headers

Save Copy Collapse All Expand All Filter JSON


```
▼ 0:
    last_name:      "Smith"
    first_name:     "John"
    department:     "IT"
▼ 1:
    last_name:      "Garcia"
    first_name:     "Maria"
    department:     "Human Resources"
▼ 2:
    last_name:      "Lopez"
    first_name:     "David"
    department:     "Sales"
▼ 3:
    last_name:      "Cruz"
    first_name:     "Jessica"
    department:     "Finance"
```

J. TRY THE AGGREGATES METHODS IN EmployeesController

